# VL-Rethinker: Incentivizing Self-Reflection of Vision-Language Models with Reinforcement Learning

**Taro Yamada**

## Abstract

Recent progress in long-chain reasoning has been driven by reinforcement learning, yet multimodal reasoning still lags behind text-only systems. We study why direct group-relative reinforcement learning is brittle for vision-language models and identify a simple failure mode: as binary rewards saturate within a prompt group, the resulting advantages collapse toward zero, sharply reducing useful policy updates and discouraging reflective behavior. We propose `VL-Rethinker`, a distillation-free framework that addresses this issue with two coupled components. First, Selective Sample Replay (SSR) keeps and replays only informative trajectories with non-zero advantages, prioritizing them by advantage magnitude to preserve learning signal near decision boundaries. Second, Forced Rethinking (FoRet) explicitly trains self-verification by triggering a second-pass continuation on a subset of rollouts and applying an auxiliary imitation loss only to successful reflective traces. `VL-Rethinker` is trained from Qwen2.5-VL-Instruct backbones on approximately 38.9K curated, verifiable multimodal queries. On the 72B scale, it reaches 80.4 on MathVista, 63.5 on MathVerse, 44.9 on MathVision, 55.9 on MMMU-Pro, 38.5 on EMMA, and 51.3 on MEGA core, while remaining below the strongest open model on MMMU val. On the 7B scale, it achieves 74.9, 54.2, 32.3, and 41.7 on MathVista, MathVerse, MathVision, and MMMU-Pro, respectively, outperforming prior open-source reasoning VLMs. Ablations show that both SSR and Forced Rethinking are necessary, with SSR improving training stability and Forced Rethinking yielding the largest gains on hard visual math benchmarks.

## 1 Introduction

Vision-language models (VLMs) have advanced rapidly on perception-heavy tasks, aided by large-scale instruction tuning and improved multimodal architectures [Dai et al., 2023, Liu et al., 2023, Li et al., 2024, Chen et al., 2024a, Hurst et al., 2024, Team et al., 2023, Anthropic, 2024]. However, robust *slow-thinking* multimodal reasoning remains notably weaker than its text-only counterpart. Recent language-model work shows that reinforcement learning (RL) can elicit long-form reasoning, self-correction, and verification without dense human annotations [Guo et al., 2025, Jaech et al., 2024, Team et al., 2025, Muennighoff et al., 2025, Yu et al., 2025, Hu et al., 2025, Zeng et al., 2025]. In multimodal settings, by contrast, gains have been smaller and less stable, even for methods explicitly optimized for reasoning [Zhang et al., 2025, Huang et al., 2025, Peng et al., 2025, Meng et al., 2025, Yang et al., 2025, Deng et al., 2025].

A natural hypothesis is that multimodal RL simply lacks sufficiently large and clean data. Data quality does matter, but our experiments and code-level analysis point to a more basic optimization issue. In group-relative policy learning with mostly binary rewards, once a prompt group becomes nearly all-correct or all-incorrect, the within-group reward differences shrink, so the corresponding advantages become zero or nearly zero. This causes a large fraction of sampled trajectories to contribute little gradient signal. The effect is particularly harmful in multimodal reasoning because

Preprint. Under review.

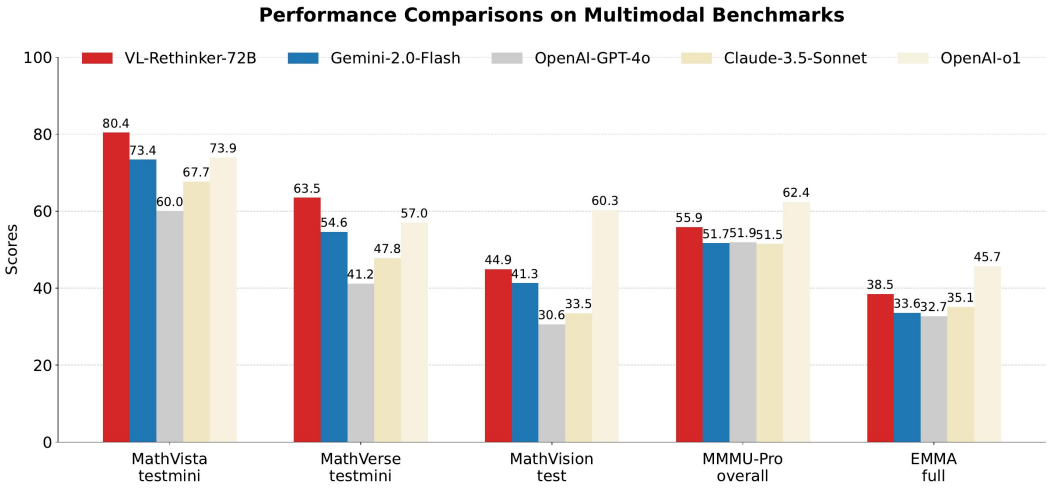



Figure 1: Performance comparison between `VL-Rethinker` and other state-of-the-art models on multimodal reasoning benchmarks. The largest gains appear on visual math and general multimodal reasoning benchmarks, while MMMU val remains a comparatively harder setting.

the policy must jointly improve perception, symbolic manipulation, and error recovery, all under sparse verification signals.

This paper introduces `VL-Rethinker`, a direct RL framework for reflective VLM reasoning. The method has two components. The first is *Selective Sample Replay* (SSR), which replays only informative samples whose advantages remain non-zero and prioritizes them by magnitude, thereby focusing updates on decision-boundary cases instead of saturated groups. The second is *Forced Rethinking*, which explicitly induces self-verification by appending short trigger phrases such as "Wait" or "Wait, does it seem right?" to a subset of ongoing rollouts and then supervising successful reflective continuations through an auxiliary imitation term. Unlike distillation-heavy pipelines, this mechanism uses the model's own trajectories and rule-based rewards.

Our study is motivated by two questions. First, can direct RL for VLMs be stabilized when group-relative advantages vanish? Second, can reflective behaviors such as self-questioning and self-correction be *trained*, rather than merely prompted at test time? The empirical answer to both is yes. Across visual mathematics, expert multi-discipline reasoning, and real-world multimodal reasoning benchmarks [Lu et al., 2023, Zhang et al., 2024, Wang et al., 2024, Yue et al., 2024, Hao et al., 2025, Chen et al., 2024b], `VL-Rethinker` sets new open-source results at 7B and 72B scale, with consistent gains over recent reasoning-oriented VLMs. At the same time, one benchmark in our evaluation, MMMU val, remains below the best open result, which we report transparently.

Our contributions are fourfold:

- We identify *vanishing advantages* as a practical failure mode of group-relative RL for VLMs with mostly binary rewards.
- We propose SSR, a replay strategy that retains and prioritizes non-zero-advantage trajectories to restore usable policy gradients.
- We introduce Forced Rethinking, a teacher-free mechanism that explicitly trains self-verification and selective correction.
- We obtain strong open-source results across multiple multimodal reasoning benchmarks at both 7B and 72B scales.

## 2 Related Work

**Reasoning-oriented RL for language models.** Recent work shows that rule-based RL can induce strong reasoning behavior in language models, including long chain-of-thought, verification, and improved test-time scaling [Guo et al., 2025, Jaech et al., 2024, Team et al., 2025, Muennighoff

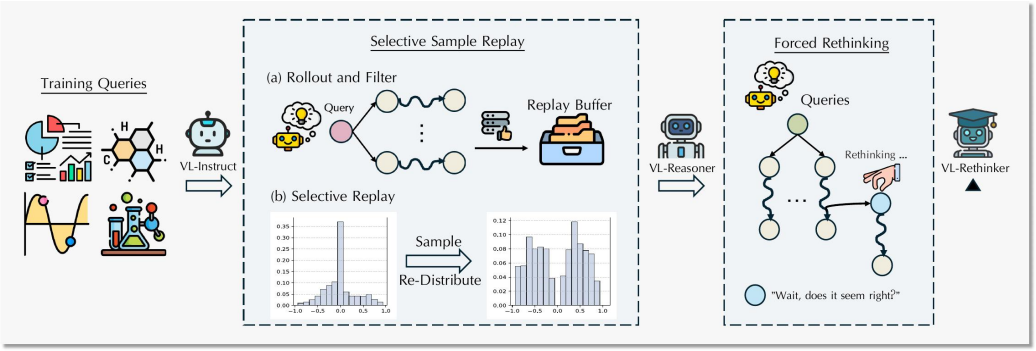



Figure 2: **Method Overview.** `VL-Rethinker` combines group-relative policy learning with two additions: SSR, which keeps informative non-zero-advantage trajectories in replay, and Forced Rethinking, which appends short self-check triggers to selected rollouts and imitates successful reflective continuations.

et al., 2025, Yu et al., 2025, Hu et al., 2025, Zeng et al., 2025]. Several studies also analyze optimization pathologies in group-based RL, including response-length bias and instability in zero-RL training [Liu et al., 2025, Zeng et al., 2025]. Our work is aligned with this line of research but focuses on a multimodal failure mode: in VLM training with verifiable binary rewards, group-relative advantages frequently collapse to zero, making naive reuse of text-only recipes ineffective.

**Multimodal reasoning models.** Open-source VLMs have improved steadily through better backbones, instruction tuning, and reasoning-focused data [Dai et al., 2023, Liu et al., 2023, Li et al., 2024, Chen et al., 2024a, Guo et al., 2024, Jia et al., 2025]. More recent reasoning-oriented models include OpenVLThinker, R1-OneVision, Vision-R1, LMM-R1, R1-VL, and MM-Eureka [Deng et al., 2025, Yang et al., 2025, Huang et al., 2025, Peng et al., 2025, Zhang et al., 2025, Meng et al., 2025]. Most of these approaches rely on strong supervision, synthetic rationales, staged text-to-vision transfer, or dense process rewards. In contrast, `VL-Rethinker` emphasizes *direct*, rule-based multimodal RL with limited additional machinery: no teacher distillation, no separate process reward model, and no multimodal-to-text formalization stage.

**Self-verification and reflection.** Self-correction has been studied in language reasoning through process supervision, verification, or budget forcing [Lightman et al., 2023, Muennighoff et al., 2025]. In multimodal models, reflection is often encouraged through prompting or distilled traces, but remains difficult to learn reliably through RL alone [Huang et al., 2025, Yang et al., 2025]. Our Forced Rethinking module turns reflection into an explicit training-time intervention: the policy is occasionally required to continue from a self-check trigger, and only successful reflective continuations receive auxiliary imitation signal.

**Multimodal reasoning benchmarks.** We evaluate on visual mathematics and multi-discipline reasoning datasets that require nontrivial interaction between perception and reasoning, including MathVista [Lu et al., 2023], MathVerse [Zhang et al., 2024], MathVision [Wang et al., 2024], MMMU [Yue et al., 2024], EMMA [Hao et al., 2025], and MEGA-Bench [Chen et al., 2024b]. These benchmarks cover diagrams, charts, geometry, tables, documents, and real-world multimodal tasks, and they expose failure cases that are often hidden by easier perception-only evaluations.

# 3 Method

## 3.1 Problem Setup

We consider multimodal queries $x$ consisting of an image and question pair. For each query, the policy $\pi_\theta$ generates $G$ candidate responses $\{y_i\}_{i=1}^{G}$. The training data contains approximately 38.9K curated, rule-verifiable multimodal queries drawn from a broad mix of mathematical, scientific, chart, document, and general reasoning tasks. We initialize from instruction-tuned Qwen2.5-VL backbones and optimize using rule-based rewards rather than distilling reasoning traces from a stronger teacher.

Each sampled response receives a scalar reward based on final answer correctness and basic validity checks. In the implementation used for this paper, valid correct responses receive reward $1.0$, invalid responses receive a small penalty ($-0.1$), and some responses containing natural, non-excessive reflective behavior receive slight shaping bonuses or penalties. The core supervision is therefore still sparse and near-binary.

### 3.2 Vanishing Advantages in Group-Relative RL

Group-relative optimization is attractive for verifiable reasoning because it avoids training a separate reward model and naturally compares multiple responses to the same prompt. Let $r_i$ denote the reward for response $y_i$ in a group of size $G$. We use the group-centered advantage

$$a_i = r_i - \frac{1}{G}\sum_{j=1}^{G} r_j. \tag{1}$$

The policy is updated with a PPO-style clipped objective [Schulman et al., 2017] over the sampled responses,

$$\mathcal{L}_{\text{RL}} = -\mathbb{E}\left[\min\left(\rho_i a_i, \text{clip}(\rho_i, 1-\epsilon_\ell, 1+\epsilon_h) a_i\right)\right], \tag{2}$$

where $\rho_i$ is the ratio between current and behavior policy probabilities for the sampled tokens.

This estimator is effective only when responses within a group differ in reward. If all samples in a group are correct, or all are wrong, then $a_i = 0$ for every $i$ and the group contributes essentially no policy-learning signal. With sparse binary rewards, such groups become increasingly common as training improves on easier queries and remains uniformly poor on the hardest ones. Figure 3 illustrates this saturation effect.

This phenomenon differs from ordinary high-variance policy gradients. The issue is not merely noisy rewards, but the *absence* of informative pairwise contrast inside a group. In practice, we observe that the replay buffer becomes dominated by zero-advantage samples unless we filter them explicitly.

### 3.3 Selective Sample Replay

To counter this collapse, we introduce SSR. The replay buffer stores trajectories after reward computation, but unlike uniform replay or unfiltered minibatch reuse, it retains only samples whose final advantages are non-zero. Intuitively, these are the responses that still lie near the model's current decision boundary, where policy changes can alter the relative ordering of outcomes.

Formally, let $\mathcal{B}_t$ denote the current set of post-rollout samples at iteration $t$ and let

$$\mathcal{I}_t = \{i \in \mathcal{B}_t : |a_i| > 0\}. \tag{3}$$

SSR forms the training minibatch from a mixture of current informative samples and previously retained informative samples. Sampling probability is proportional to

$$p_i \propto (|a_i| + \delta)^\alpha, \tag{4}$$

with a small $\delta > 0$ for numerical stability. This design is inspired by prioritized replay [Schaul et al., 2015], but adapted to near-on-policy language-model RL: the buffer keeps only recent informative trajectories, and replay is used to densify learning signal rather than to learn a fully off-policy value function.

The implementation uses a persistent pool of previously informative samples, plus a current-batch subset. When SSR is enabled, training samples are drawn from both sources, weighted by absolute advantage magnitude. This produces a curriculum-like effect: trivial saturated examples are down-weighted, while partially solved or borderline prompts remain active longer.

### 3.4 Forced Rethinking

Zero-shot prompting can ask a VLM to "double check" an answer, but such behavior is often inconsistent and wasteful. We instead *train* selective reflection. For a subset of sampled trajectories, after the first-pass response $y^{(0)}$ is produced, we append a short trigger $\tau$ chosen from a small set:

$$\begin{aligned}\tau \in \{&\texttt{Wait}, \texttt{Wait, does it seem right?},\\ &\texttt{Wait, let's double check}, \texttt{Wait, there might be a mistake}\}.\end{aligned} \tag{5}$$

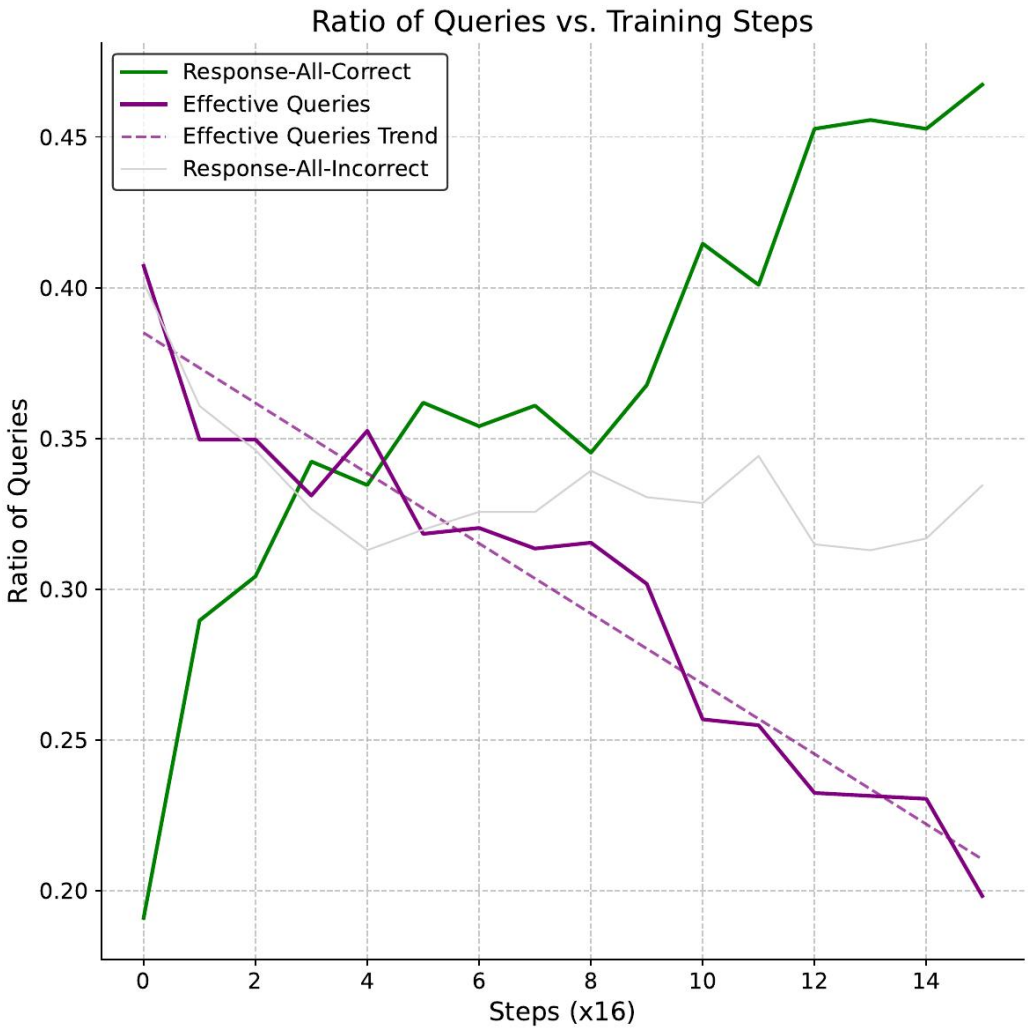



Figure 3: Illustration of the Vanishing Advantages problem. When a prompt group becomes uniformly correct or uniformly incorrect, group-relative advantages collapse toward zero and policy updates weaken.

The model then generates a second continuation $y^{(1)}$ conditioned on the original multimodal prompt and the partial first-pass answer plus the trigger.

The trigger policy is selective rather than uniform. In training, rethinking is mainly applied to groups that are not already saturated and to samples that are still informative under the current reward signal. The code path also stochastically subsamples eligible trajectories, preventing the model from learning to always overthink.

Forced Rethinking serves two roles. First, it expands exploration into reflective continuations that the base policy might rarely enter on its own. Second, when a rethinking trajectory contains reflective markers and obtains positive advantage, we add an auxiliary imitation term

$$\mathcal{L}_{\text{aux}} = -\sum_{t \in y^{(1)}} \log \pi_\theta(y_t^{(1)} \mid x, y_{<t}^{(1)}), \tag{6}$$

scaled by coefficient $\lambda$. In our implementation, this auxiliary loss is activated only for successful reflective traces, making it teacher-free: the model learns from its own good rethinking trajectories rather than from externally distilled chains.

The final training objective is

$$\mathcal{L} = \mathcal{L}_{\text{RL}} + \lambda \mathcal{L}_{\text{aux}}. \tag{7}$$

In the released training scripts, $\lambda = 0.05$ and KL regularization is disabled, reflecting the near-binary, verifiable-reward regime used throughout our experiments.

### 3.5 Why the Combination Matters

SSR and Forced Rethinking target complementary failure modes. SSR restores informative gradients by concentrating updates on samples with non-zero advantages. Forced Rethinking improves the policy's action space by increasing the chance that a trajectory enters a self-checking or corrective mode. Without SSR, the second component lacks stable optimization signal; without Forced Rethinking, the replayed trajectories improve answer selection but do not reliably teach the model when or how to re-evaluate an answer. The ablations in Section 6 support this interaction.

## 4 Experimental Setup

**Training data.** Our main models are trained on ViRL39K, a curated collection of approximately 38.9K verifiable multimodal reasoning queries spanning visual mathematics, science, charts, tables, documents, spatial reasoning, and broader STEM or social-science topics. The 7B and 72B models are initialized from Qwen2.5-VL-Instruct checkpoints, and we also report a 32B model in the 72B-scale comparison table to show scaling trends. For controlled algorithmic ablations, we use a 16K subset and additional reduced variants that remove text-only and science-plus-text subsets, matching Table 3.

**Training protocol.** Training uses the OpenRLHF-based implementation released with the project. The scripts sample $8$ responses per prompt, use group-based advantages, no KL penalty, an auxiliary coefficient of $0.05$, and generation temperatures of $1.0$ for training and $0.6$ for evaluation. The method is near on-policy: replay is limited to recent informative trajectories and used only for actor updates.

**Evaluation.** We report results on MathVista testmini [Lu et al., 2023], MathVerse testmini [Zhang et al., 2024], MathVision test [Wang et al., 2024], MMMU-Pro overall and MMMU val [Yue et al., 2024], EMMA full [Hao et al., 2025], and MEGA core [Chen et al., 2024b]. Following the provided evaluation code, correctness is measured with rule-based judgment functions aligned with the training reward where possible. We compare against both general-purpose VLMs and reasoning-specialized multimodal models.

## 5 Results

### 5.1 Main Results at 72B Scale

Table 1 shows that `VL-Rethinker`-72B outperforms the strongest open-source baselines on six of seven reported benchmarks. The gains are largest on visual mathematics: 80.4 on MathVista, 63.5 on MathVerse, and 44.9 on MathVision, improving over the previous open best by 5.6, 6.3, and 6.8 points, respectively. The method also improves over prior open models on MMMU-Pro, EMMA, and MEGA core by 3.7, 4.4, and 2.3 points.

The main exception is MMMU val, where `VL-Rethinker`-72B reaches 68.8, trailing the best open result by 1.4 points. This is important because it shows the method is not uniformly dominant across all multimodal reasoning distributions. Our gains are strongest on benchmarks with explicit answer verification and substantial visual-mathematical structure; more heterogeneous expert-domain evaluation remains challenging.

The 32B model follows a similar trend, exceeding the open baselines on MathVista, MathVision, and EMMA but lagging behind the 72B model overall. This suggests the benefits of SSR and rethinking scale with model size, but larger models still matter for full cross-domain generalization.

### 5.2 Main Results at 7B Scale

At 7B, `VL-Rethinker` remains consistently competitive and often substantially stronger than previous open-source reasoning VLMs. It reaches 74.9 on MathVista, 54.2 on MathVerse, 32.3 on MathVision,

| Model | Math-Related | | | Multi-Discipline | | | Real-World |
|---|---|---|---|---|---|---|---|
| | MathVista testmini | MathVerse testmini | MathVision test | MMMU-Pro overall | MMMU val | EMMA full | MEGA core |
| Proprietary Model | | | | | | | |
| OpenAI-o1 | 73.9 | 57.0 | 60.3 | 62.4 | 78.2 | 45.7 | 56.2 |
| OpenAI-GPT-4o | 60.0 | 41.2 | 30.6 | 51.9 | 69.1 | 32.7 | 52.7 |
| Claude-3.5-Sonnet | 67.7 | 47.8 | 33.5 | 51.5 | 68.3 | 35.1 | 52.3 |
| Gemini-2.0-Flash | 73.4 | 54.6 | 41.3 | 51.7 | 70.7 | 33.6 | 54.1 |
| Open-Source Models | | | | | | | |
| Llama4-Scout-109B | 70.7 | - | - | 52.2 | 69.4 | 24.6 | 31.8 |
| InternVL-2.5-78B | 72.3 | 51.7 | 34.9 | 48.6 | 61.8 | 27.1 | 44.1 |
| QvQ-72B | 71.4 | 48.6 | 35.9 | 51.5 | 70.3 | 32.0 | 8.8 |
| LLava-OV-72B | 67.5 | 39.1 | 30.1 | 31.0 | 56.8 | 23.8 | 29.7 |
| Qwen-2.5-VL-32B | 74.7 | 48.5 | 38.4 | 49.5 | †59.4 | 31.1 | 13.3 |
| Qwen-2.5-VL-72B | 74.8 | 57.2 | 38.1 | 51.6 | †67.0 | 34.1 | 49.0 |
| `VL-Rethinker-32B` | 78.8 | 56.9 | 40.5 | 50.6 | 65.6 | 37.9 | 19.9 |
| `VL-Rethinker-72B` | **80.4** | **63.5** | **44.9** | **55.9** | **68.8** | **38.5** | **51.3** |
| Δ (Ours - Open SoTA) | +5.6 | +6.3 | +6.8 | +3.7 | -1.4 | +4.4 | +2.3 |

Table 1: Comparison between our 72B model and other state-of-the-art models. The notation of † indicates reproduced results using our evaluation protocols.

| Model | Math-Related | | | Multi-Discipline | | | Real-World |
|---|---|---|---|---|---|---|---|
| | MathVista testmini | MathVerse testmini | MathVision test | MMMU-Pro overall | MMMU val | EMMA full | MEGA core |
| General Vision-Language Models | | | | | | | |
| InternVL2-8B | 58.3 | - | 17.4 | 29.0 | 51.2 | 19.8 | 26.0 |
| InternVL2.5-8B | 64.4 | 39.5 | 19.7 | 34.3 | 56.0 | - | 30.4 |
| QwenVL2-7B | 58.2 | - | 16.3 | 30.5 | 54.1 | 20.2 | 34.8 |
| QwenVL2.5-7B | 68.2 | 46.3 | 25.1 | 36.9 | †54.3 | 21.5 | 35.0 |
| Llava-OV-7B | 63.2 | 26.2 | - | 24.1 | 48.8 | 18.3 | 22.9 |
| Kimi-VL-16B | 68.7 | 44.9 | 21.4 | - | †55.7 | - | - |
| Vision-Language Reasoning Models | | | | | | | |
| MM-Eureka-8B (Intern) | 67.1 | 40.4 | 22.2 | 27.8 | 49.2 | - | - |
| MM-Eureka-7B (Qwen) | 73.0 | 50.3 | 26.9 | - | - | - | - |
| R1-VL-7B | 63.5 | 40.0 | 24.7 | 7.8 | 44.5 | 8.3 | 29.9 |
| R1-Onevision-7B | 64.1 | 46.4 | 29.9 | 21.6 | - | 20.8 | 27.1 |
| OpenVLThinker-7B | 70.2 | 47.9 | 25.3 | 37.3 | 52.5 | 26.6 | 12.0 |
| `VL-Rethinker-7B` | **74.9** | **54.2** | **32.3** | **41.7** | **56.7** | **29.7** | **37.2** |
| Δ (Ours - Prev SoTA) | +4.7 | +6.3 | +2.4 | +4.4 | +0.7 | +3.1 | +2.2 |

Table 2: Comparison between our 7B model and other general and reasoning vision-language models. † means that the results are reproduced by us.

41.7 on MMMU-Pro, 56.7 on MMMU val, 29.7 on EMMA, and 37.2 on MEGA core. Relative to the previous best open models, the improvements range from 0.7 to 6.3 points.

Two observations stand out. First, the method is not restricted to visual mathematics: MMMU-Pro, EMMA, and MEGA core all improve as well. Second, the 7B gains on MathVerse and MathVision are especially large, which is consistent with our hypothesis that explicit self-verification is particularly useful when diagrams and symbolic reasoning interact.

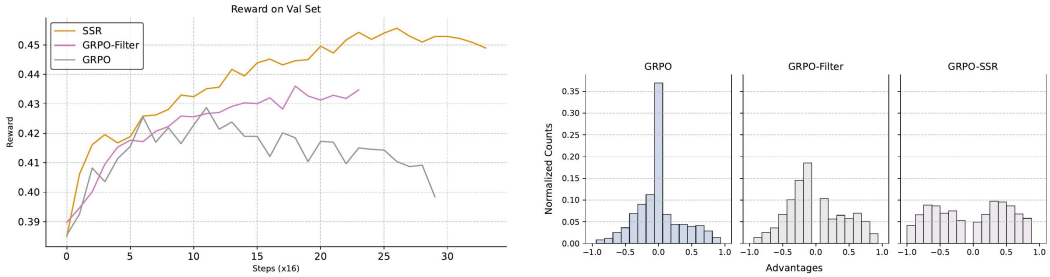
# 6 Analysis

## 6.1 Ablation on SSR and Forced Rethinking

Table 3 isolates the effect of the two main components on the 7B model. The full system achieves 32.3 on MathVision, 74.9 on MathVista, 54.2 on MathVerse, 41.7 on MMMU-Pro, and 29.7 on EMMA. Removing Forced Rethinking while keeping SSR drops performance to 29.8, 72.4, 53.2, 40.9, and 29.5, respectively. Replacing SSR with a simple filter lowers results further to 28.5, 72.0,

| Model | RL-Algo | Data | MathVision | MathVista | MathVerse | MMMU-Pro | EMMA |
|---|---|---|---|---|---|---|---|
| VL-Rethinker-7B | SSR | 16K | 32.3 | 74.9 | 54.2 | 41.7 | 29.7 |
| w/o 'Forced-Rethinking' | SSR | 16K | 29.8 | 72.4 | 53.2 | 40.9 | 29.5 |
| - no SSR | Filter | 16K | 28.5 | 72.0 | 50.0 | 40.0 | 26.9 |
| - no SSR& Filter | GRPO | 16K | 26.0 | 70.9 | 51.4 | 38.8 | 26.2 |
| - no Text | SSR | 13K | 29.1 | 73.5 | 53.5 | 41.1 | 28.7 |
| - no Science&Text | SSR | 11K | 28.0 | 71.6 | 50.3 | 39.7 | 28.0 |

Table 3: Ablation Results to show the impact of SSR and Data Mix.




(a) Evaluation curves under GRPO, simple filtering, and SSR.

(b) Advantage distribution shifts away from zero under SSR.

Figure 4: Comparisons of training dynamics of GRPO, GRPO-Filter and GRPO-SSR. SSR both stabilizes evaluation performance and preserves a larger mass of informative non-zero advantages.

50.0, 40.0, and 26.9. Using vanilla GRPO without filtering or SSR performs worst overall, especially on MathVision, where performance falls to 26.0.

The largest absolute degradation appears on MathVision: 32.3 $\rightarrow$ 29.8 without Forced Rethinking and 32.3 $\rightarrow$ 26.0 without both components. This benchmark contains competition-style visual math problems that strongly benefit from explicit verification and from retaining informative decision-boundary samples. The results indicate that SSR and rethinking are not redundant; they contribute distinct gains.

The same table also studies data mix. Removing text data reduces MathVision from 32.3 to 29.1 and MathVista from 74.9 to 73.5, while removing both science and text data lowers them further to 28.0 and 71.6. These drops suggest that although the proposed RL algorithm is central, training data diversity remains important for transferring reflective reasoning beyond the narrow set of visual math tasks.

### 6.2 Training Dynamics

Figure 4 visualizes the optimization behavior underlying the ablation results. Relative to vanilla GRPO and simple filtering, SSR yields more stable evaluation curves and a broader advantage distribution that does not collapse around zero as quickly. This is exactly the intended behavior: by keeping non-zero-advantage samples active, replay prevents training from being dominated by saturated groups.

The comparison between simple filtering and SSR is informative. Filtering discards zero-advantage samples in the current batch, but it does not retain a memory of previously informative cases. As a result, once the current on-policy batch becomes too easy or too uniformly hard, learning signal still deteriorates. SSR addresses this by mixing current informative samples with replayed informative ones, which is why it improves both convergence and final accuracy.

### 6.3 Effect of Rethinking Strategy

Figure 5 compares rethinking strategies. The main lesson is that reflection should be *selective*. Uniformly encouraging extra deliberation is not the goal; instead, the model should learn to revisit an answer when uncertainty or mismatch remains. The trigger-based second pass offers a simple way to

expose the policy to such states during RL. The auxiliary imitation term then preserves successful reflective patterns that would otherwise be too sparse to learn reliably from the policy gradient alone.

This interpretation is consistent with the implementation. Trigger phrases are appended only to eligible trajectories, and successful reflective continuations are supervised only when they also carry positive advantage. In other words, the method rewards useful rethinking rather than verbosity for its own sake.

### 6.4 Discussion

The strongest gains of `VL-Rethinker` occur on benchmarks where answer verification is reliable and wrong intermediate reasoning can plausibly be corrected by a second look at the image or at the symbolic argument. This includes MathVista, MathVerse, and MathVision. The smaller but still positive gains on MMMU-Pro, EMMA, and MEGA indicate that the benefits extend beyond narrow visual math. However, the weaker MMMU val result suggests that explicit self-reflection alone does not solve all multimodal reasoning failures, especially when domain diversity and evaluation heterogeneity are high.

A complementary way to interpret these results is through problem difficulty. Easy prompts quickly become all-correct, so vanilla group-relative RL spends many updates on saturated groups. Extremely hard prompts often remain all-wrong, which is equally uninformative. The useful learning regime lies between these extremes, where some responses are correct, others are not, and a second-pass check can still change the outcome. SSR keeps these borderline trajectories active after they would disappear from the current batch, while Forced Rethinking increases the chance that a borderline but recoverable trajectory turns into a successful correction. This perspective helps explain why the largest gains appear on multi-step visual math benchmarks and why improvements are smaller on broader expert-domain evaluations.

### 6.5 Where the Gains Come From

The benchmark pattern above suggests three concrete effects that explain both the gains and their scope:

- SSR reallocates updates away from saturated prompts and keeps mixed-outcome prompts active longer.
- Forced Rethinking exposes recovery trajectories in which re-reading the image or recomputing a symbolic step flips the final reward.
- The gains therefore concentrate on recoverable reasoning errors, not on trivial prompts or clearly out-of-distribution failures.

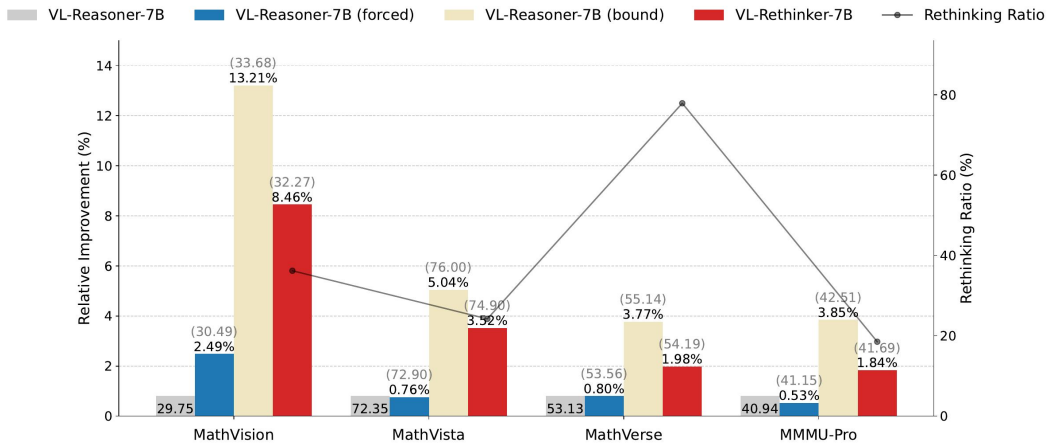



Figure 5: **Relative Improvement with Different Re-thinking Strategies.** Training-time rethinking provides the strongest gains when triggers are short, selective, and coupled with RL rather than used as a universal prompt prefix.

### 6.6 Limits of the Mechanism

When an entire group is uniformly wrong because the query is genuinely out of distribution, replay cannot create contrast that is absent from the sampled rollouts, and rethinking has no successful correction to imitate.

The results imply a narrow success regime for the method:

- **Reachable errors.** If the base policy can already produce some correct samples, SSR preserves those mixed-outcome groups instead of letting them disappear once batches saturate.
- **Recoverable errors.** If a wrong first pass can be fixed by re-reading the image or recomputing a step, Forced Rethinking creates the exploration needed for that correction to appear and be imitated.
- **Irrecoverable errors.** If evidence is ambiguous, verification is noisy, or the model lacks the needed knowledge, neither replay nor rethinking can manufacture a correct trajectory.
- **Budget allocation.** RL budget is best spent on prompts that are neither already solved nor hopeless.

This boundary matches the stronger gains on verifiable visual reasoning and the smaller gains on broader heterogeneous benchmarks.

Overall, the analysis supports a simple picture: group-relative RL for VLMs fails when too many samples become locally uninformative, and reflective behavior does not emerge reliably unless exploration is steered toward it. SSR fixes the first issue; Forced Rethinking addresses the second.

## 7 Conclusion

We presented VL-Rethinker, a distillation-free framework for multimodal reasoning that combines Selective Sample Replay with Forced Rethinking. The method is motivated by a concrete optimization diagnosis: under sparse binary rewards, group-relative advantages for VLMs often vanish, starving policy learning and suppressing self-correction. SSR restores informative gradients by replaying non-zero-advantage samples, while Forced Rethinking explicitly trains self-verification and correction through trigger-based second-pass rollouts and auxiliary imitation on successful traces.

Empirically, VL-Rethinker establishes strong open-source results across a diverse set of multimodal reasoning benchmarks at 7B and 72B scales, with especially large improvements on visual mathematics. The results also clarify the method's limits: although it improves most benchmarks, MMMU val remains challenging. Taken together, the findings show that reflective multimodal reasoning can be improved substantially through direct RL, provided that the optimization signal is preserved and self-reflection is trained explicitly rather than assumed to emerge automatically. More broadly, the results point to a practical recipe: preserve informative contrasts, then use them on trajectories where reflection can still change the answer.

## References


Wenliang Dai, Junnan Li, Dongxu Li, Anthony Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale Fung, and Steven Hoi. InstructBLIP: Towards general-purpose vision-language models with instruction tuning. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023. URL https://openreview.net/forum?id=vvoWPYqZJA.

Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. *Advances in neural information processing systems*, 36:34892–34916, 2023.

Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Peiyuan Zhang, Yanwei Li, Ziwei Liu, et al. Llava-onevision: Easy visual task transfer. *arXiv preprint arXiv:2408.03326*, 2024.

Zhe Chen, Weiyun Wang, Yue Cao, Yangzhou Liu, Zhangwei Gao, Erfei Cui, Jinguo Zhu, Shenglong Ye, Hao Tian, Zhaoyang Liu, et al. Expanding performance boundaries of open-source multimodal models with model, data, and test-time scaling. *arXiv preprint arXiv:2412.05271*, 2024a.

Aaron Hurst, Adam Lerer, Adam P Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, et al. Gpt-4o system card. *arXiv preprint arXiv:2410.21276*, 2024.

Gemini Team, Rohan Anil, Sebastian Borgeaud, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, Katie Millican, et al. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2023.

Anthropic. Claude 3.5 sonnet model card addendum, 2024. URL https://www.anthropic.com/claude-3-5-sonnet-model-card-addendum.

Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.

Aaron Jaech, Adam Kalai, Adam Lerer, Adam Richardson, Ahmed El-Kishky, Aiden Low, Alec Helyar, Aleksander Madry, Alex Beutel, Alex Carney, et al. Openai o1 system card. *arXiv preprint arXiv:2412.16720*, 2024.

Kimi Team, Angang Du, Bofei Gao, Bowei Xing, Changjiu Jiang, Cheng Chen, Cheng Li, Chenjun Xiao, Chenzhuang Du, Chonghua Liao, et al. Kimi k1. 5: Scaling reinforcement learning with llms. *arXiv preprint arXiv:2501.12599*, 2025.

Niklas Muennighoff, Zitong Yang, Weijia Shi, Xiang Lisa Li, Li Fei-Fei, Hannaneh Hajishirzi, Luke Zettlemoyer, Percy Liang, Emmanuel Candès, and Tatsunori Hashimoto. s1: Simple test-time scaling. *arXiv preprint arXiv:2501.19393*, 2025.

Qiying Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Tiantian Fan, Gaohong Liu, Lingjun Liu, Xin Liu, et al. Dapo: An open-source llm reinforcement learning system at scale. *arXiv preprint arXiv:2503.14476*, 2025.

Jingcheng Hu, Yinmin Zhang, Qi Han, Daxin Jiang, Xiangyu Zhang, and Heung-Yeung Shum. Open-reasoner-zero: An open source approach to scaling up reinforcement learning on the base model. *arXiv preprint arXiv:2503.24290*, 2025.

Weihao Zeng, Yuzhen Huang, Qian Liu, Wei Liu, Keqing He, Zejun Ma, and Junxian He. Simplerl-zoo: Investigating and taming zero reinforcement learning for open base models in the wild. *arXiv preprint arXiv:2503.18892*, 2025.

Jingyi Zhang, Jiaxing Huang, Huanjin Yao, Shunyu Liu, Xikun Zhang, Shijian Lu, and Dacheng Tao. R1-vl: Learning to reason with multimodal large language models via step-wise group relative policy optimization. *arXiv preprint arXiv:2503.12937*, 2025.

Wenxuan Huang, Bohan Jia, Zijie Zhai, Shaosheng Cao, Zheyu Ye, Fei Zhao, Yao Hu, and Shaohui Lin. Vision-r1: Incentivizing reasoning capability in multimodal large language models. *arXiv preprint arXiv:2503.06749*, 2025.

Yingzhe Peng, Gongrui Zhang, Miaosen Zhang, Zhiyuan You, Jie Liu, Qipeng Zhu, Kai Yang, Xingzhong Xu, Xin Geng, and Xu Yang. Lmm-r1: Empowering 3b lmms with strong reasoning abilities through two-stage rule-based rl. *arXiv preprint arXiv:2503.07536*, 2025.

Fanqing Meng, Lingxiao Du, Zongkai Liu, Zhixiang Zhou, Quanfeng Lu, Daocheng Fu, Botian Shi, Wenhai Wang, Junjun He, Kaipeng Zhang, et al. Mm-eureka: Exploring visual aha moment with rule-based large-scale reinforcement learning. *arXiv preprint arXiv:2503.07365*, 2025.

Yi Yang, Xiaoxuan He, Hongkun Pan, Xiyan Jiang, Yan Deng, Xingtao Yang, Haoyu Lu, Dacheng Yin, Fengyun Rao, Minfeng Zhu, et al. R1-onevision: Advancing generalized multimodal reasoning through cross-modal formalization. *arXiv preprint arXiv:2503.10615*, 2025.

Yihe Deng, Hritik Bansal, Fan Yin, Nanyun Peng, Wei Wang, and Kai-Wei Chang. Openvlthinker: An early exploration to complex vision-language reasoning via iterative self-improvement. *arXiv preprint arXiv:2503.17352*, 2025.

Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chunyuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-Wei Chang, Michel Galley, and Jianfeng Gao. Mathvista: Evaluating mathematical reasoning of foundation models in visual contexts. *arXiv preprint arXiv:2310.02255*, 2023.

Renrui Zhang, Dongzhi Jiang, Yichi Zhang, Haokun Lin, Ziyu Guo, Pengshuo Qiu, Aojun Zhou, Pan Lu, Kai-Wei Chang, Yu Qiao, et al. Mathverse: Does your multi-modal llm truly see the diagrams in visual math problems? In *European Conference on Computer Vision*, pages 169–186, 2024.

Ke Wang, Junting Pan, Weikang Shi, Zimu Lu, Houxing Ren, Aojun Zhou, Mingjie Zhan, and Hongsheng Li. Measuring multimodal mathematical reasoning with math-vision dataset. *Advances in Neural Information Processing Systems*, 37:95095–95169, 2024.

Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, et al. Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9556–9567, 2024.

Yunzhuo Hao, Jiawei Gu, Huichen Will Wang, Linjie Li, Zhengyuan Yang, Lijuan Wang, and Yu Cheng. Can mllms reason in multimodality? emma: An enhanced multimodal reasoning benchmark. *arXiv preprint arXiv:2501.05444*, 2025.

Jiacheng Chen, Tianhao Liang, Sherman Siu, Zhengqing Wang, Kai Wang, Yubo Wang, Yuansheng Ni, Wang Zhu, Ziyan Jiang, Bohan Lyu, et al. Mega-bench: Scaling multimodal evaluation to over 500 real-world tasks. *arXiv preprint arXiv:2410.10563*, 2024b.

Zichen Liu, Changyu Chen, Wenjun Li, Penghui Qi, Tianyu Pang, Chao Du, Wee Sun Lee, and Min Lin. Understanding r1-zero-like training: A critical perspective. *arXiv preprint arXiv:2503.20783*, 2025.

Jarvis Guo, Tuney Zheng, Yuelin Bai, Bo Li, Yubo Wang, King Zhu, Yizhi Li, Graham Neubig, Wenhu Chen, and Xiang Yue. Mammoth-vl: Eliciting multimodal reasoning with instruction tuning at scale. *arXiv preprint arXiv:2412.05237*, 2024.

Yiming Jia, Jiachen Li, Xiang Yue, Bo Li, Ping Nie, Kai Zou, and Wenhu Chen. Visualwebinstruct: Scaling up multimodal instruction data through web search. *arXiv preprint arXiv:2503.10582*, 2025.

Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step. In *The Twelfth International Conference on Learning Representations*, 2023.

John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

Tom Schaul, John Quan, Ioannis Antonoglou, and David Silver. Prioritized experience replay. *arXiv preprint arXiv:1511.05952*, 2015.